LOVNISH VERMA

Software Developer | AI/ML Engineer & Researcher
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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer and researcher with hands-on expertise in machine learning model development, NLP, and scalable data pipelines. Proficient in Python, modern web frameworks (Flask, Django), and transformer-based LLMs (BERT, GPT, LLaMA) with working knowledge of RAG pipelines and LangChain. Proven track record contributing to national-level e-Governance AI projects at NIELIT. Published author in peer-reviewed journals on Machine Learning, Blockchain, and IoT. Passionate about leveraging Generative AI and frugal engineering to build accessible, high-impact tech solutions.

PROFESSIONAL EXPERIENCE

Project Engineer | NIELIT, Ropar, Punjab *Dec 2024 – Present*

- Architected AI Solutions: Designed and implemented AI-powered systems to advance e-Governance initiatives and support institutional research goals.
- Engineered ML Integrations: Built and deployed robust REST APIs to seamlessly integrate machine learning models, ensuring scalable and efficient workflows.
- Drove Applied Research: Collaborated with senior scientists and field experts to align theoretical AI concepts with practical, real-world applications.
- Led National Tech Training: Spearheaded capacity-building programs under the MeitY FutureSkills Prime scheme, training 800+ university students and 400+ government officials in AI/ML, Data Science, and Big Data.

Software Developer | Smiling Walls, Chandigarh *Nov 2023 – Aug 2024*

- Developed Data Pipelines: Engineered automated Python-based data pipelines using Pandas and Matplotlib for comprehensive analytics and reporting.
- Optimized Operations: Automated manual data workflows, boosting overall processing efficiency by 40%.
- Deployed Edge AI: Designed and tested edge AI solutions utilizing NVIDIA Jetson Nano and Orin for real-time data processing.

AIML Trainee | NIELIT, Ropar, Punjab *Apr 2023 – Oct 2023*

- Model Development: Contributed to the design and training of NLP, computer vision, and predictive analytics models.
- Application Integration: Successfully integrated ML models into interactive user interfaces using Tkinter, PyQt, Flask, Django, Streamlit, and Gradio.
- Digital India Initiatives: Supported AI capacity-building and technical workshops aligned with Digital India national technology goals.

EDUCATION

B.Tech, Computer Science & Engineering — Green Hills Engineering College (HPTU) *2020 – May 2023*

Diploma in Computer Engineering — Govt Polytechnic College Sundernagar (HPTECH BOARD) *2017 – Jun 2020*

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript

ML/AI Frameworks: Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib, Seaborn, NLP, Computer Vision

Generative AI & LLMs: HuggingFace, Transformers (BERT, GPT, LLaMA), LangChain, RAG Pipelines

Web Frameworks: Flask, Django, Streamlit, Gradio

Databases: MySQL, MongoDB, Vector DBs (exposure)

Tools & Cloud: Git, Jupyter Notebook, Google Colab, Docker, AWS, Google Search Console

Concepts: AI Model Deployment, REST APIs, Data Pipelines, Blockchain, Edge Computing (IoT)

CERTIFICATIONS & AWARDS

- Evaluator – National Screening Round — Smart India Hackathon 2025 (Govt. of India & AICTE) | Dec 2025 [Verify Credential]
- 2nd Runner Up (out of 1,552 projects) — Paranox 2.0 National Hackathon, TechXNinjas | Nov 2025 [Verify Credential]

- Certificate of Competency: AI on Jetson Nano — NVIDIA | Nov 2025 [Verify Credential]
- Critical Thinking in the AI Era — HP LIFE & HP Foundation | Feb 2026 [Verify Credential]
- Python Developer Certification (~300 Hours) — freeCodeCamp | Feb 2026 [Verify Credential]
- CITI Program Certification: Human Research — MIT Affiliates | Feb 2026 [Verify Credential]
- Python Essentials 1 — Cisco Networking Academy & OpenEDG Python Institute | Jan 2026 [Verify Credential]
- CyberShikshaa Fundamentals — Microsoft & DSCI [Verify Credential]
- The Complete 2022 Web Development Bootcamp — Udemy [Verify Credential]

RESEARCH PUBLICATIONS

Explainable AI for Web and Text Mining *International Journal of Digital Technologies | Jan 2026 | Lead Author*

Pioneering Advances in AI, Machine Learning, and Big Data Analytics *Book Chapter | 2026 | Lead Author*

IoT-Based Remote Control and Monitoring of Electrical Appliances *International Journal of Digital Technologies | 2025 | Lead Author*

Blockchain-Based Certificate Verification System *Lecture Notes in Networks and Systems | Jul 2024 | Lead Author*

PROJECTS

Selected projects demonstrating end-to-end AI/ML development, deployment, and full-stack engineering.

1. Brain Tumor Detection AI

Problem Solved: Assists doctors by automatically detecting brain tumors from MRI scans, reducing manual diagnosis time with 95%+ accuracy.
Tech Stack: *TensorFlow, Keras (CNN), Flask, Python, Hugging Face Spaces*
Live: lovnishverma-braintumor.hf.space

2. NIELIT Ropar AI Assistant (Fine-tuned SLM)

Problem Solved: Eliminates inaccurate answers from generic LLMs by fine-tuning a domain-specific Small Language Model (Llama-3.2-1B) on NIELIT Ropar institutional data.
Tech Stack: *Meta Llama-3.2-1B, Supervised Fine-Tuning (SFT), Gradio, Hugging Face*
Live: lovnishverma-nielit-assistant.hf.space

3. Face & Emotion Recognition Attendance System

Problem Solved: Automates attendance marking using real-time face recognition and emotion detection, eliminating manual roll calls and proxy attendance.
Tech Stack: *Python, OpenCV, Face Recognition Library, Deep Learning (CNN), Streamlit, SQLite*
Live: faceemotionielit.streamlit.app

4. SmartLane AI – Intelligent Traffic Management

Problem Solved: Optimizes traffic signal timing at 4-way intersections by detecting vehicle density and classifying emergency vehicles using AI.
Tech Stack: *YOLOv8, CNN, OpenCV, Streamlit, Python*
Live: paranox-sourcecode.hf.space

5. SecureCert – Blockchain Certificate Verification

Problem Solved: Prevents certificate forgery by storing and verifying academic/professional certificates on the Ethereum blockchain.
Tech Stack: *Python Flask, Ethereum, Solidity, Web3.js*
Live: sourcecode.pythonanywhere.com

6. Intelligent Book Recommender

Problem Solved: Helps readers discover relevant books by using NLP-based plot similarity matching on a large Goodreads dataset, with sub-1-second response time.
Tech Stack: *Sentence Transformers, NLP, Gradio, Python, Hugging Face Spaces*
Live: lovnishverma-book-recommendations.hf.space

7. Diabetes Risk Predictor

Problem Solved: Provides early diabetes risk assessment from clinical parameters, enabling preventive healthcare decisions.
Tech Stack: *Scikit-Learn (Ensemble ML), Streamlit, Pandas, Python, Hugging Face*
Live: lovnishverma-diabetes.hf.space

8. NIELIT StudentHub – Zero-OpEx Knowledge Portal

Problem Solved: Delivers a fully serverless institutional knowledge management system at $0 annual maintenance cost.

Tech Stack: *HTML/JS, Google Apps Script (Backend), Google Sheets (DB), GitHub Pages*
Live: nielitropar.github.io

9. Face Recognition Attendance System (Edge AI)

Problem Solved: Provides real-time face recognition attendance tracking with CUDA acceleration, optimized for edge devices like NVIDIA Jetson Nano.
Tech Stack: *Python, Dlib (CUDA), OpenCV (GPU), NumPy, NVIDIA Jetson Nano/Orin*
Live: github.com/lovnishverma/FaceRecognition-Attendance

10. NIELIT Smart Lab — Production IoT Home Automation System

Problem Solved: Provides a production-ready, remotely manageable IoT system for educational labs — enabling dual appliance control, live environmental monitoring, and wireless firmware updates without physical access to the device.
Tech Stack: *C++ (Arduino), ESP8266 (NodeMCU), DHT11 Sensor, Dual Relay Module, WiFiManager, REST API (JSON), OTA Firmware Updates, mDNS, EEPROM Persistence, HTTP Basic Auth*
Live: github.com/lovnishverma/nielit-smartlab

11. Certificate Automation System — Future Skills Prime (Govt. of India)

Problem Solved: Eliminated manual effort in issuing hundreds of unique certificates for government bootcamp participants (GOT-A, GOT-B & Bootcamp tracks) under the MeitY Future Skills Prime initiative — reducing issuance time from days to minutes.
Tech Stack: *Python, Django, ReportLab/FPDF (PDF generation), Mailjet API (bulk email), QR Code embedding, OpenPyXL/Pandas (Excel/CSV input)*
Live: github.com/lovnishverma/certificate-email-sender (Public Demo)

HOBBIES & INTERESTS

- **Content Creation:** Blogging on Medium.com; Social Media Content Creator
- **Social Media:** LinkedIn Networking & Community Building
- **Analytics & Data Science:** Open Source Contributor, Data Analysis, ML/DL experiments, SEO Experimentation

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